

BIOS5802: ADVANCED METHODS IN BIOSTATISTICS

Effective Term

Semester A 2025/26

Part I Course Overview

Course Title

Advanced Methods in Biostatistics

Subject Code

BIOS - Biostatistics

Course Number

5802

Academic Unit

Biostatistics (BIOS)

College/School

College of Computing (CC)

Course Duration

One Semester

Credit Units

3

Level

P5, P6 - Postgraduate Degree

Medium of Instruction

English

Medium of Assessment

English

Prerequisites

Nil

Precursors

Nil

Equivalent Courses

Nil

Exclusive Courses

Nil

Part II Course Details

Abstract

This course aims to provide students with the tools to carry out regression analyses involving non-continuous (e.g., binary or count-valued) responses, correlated observations, and censored data. Topics covered include: generalized linear models (GLM) for non-Gaussian response, mixed-effects models and generalized estimating equations (GEE) for correlated observations, longitudinal data analysis, and Cox proportional hazards models for the analysis of survival time outcomes. Examples are drawn from public health and the biomedical sciences.

Course Intended Learning Outcomes (CILOs)

CILOs		Weighting (if app.)	DEC-A1	DEC-A2	DEC-A3
1	Develop and fit regression models for discrete responses	40	x	x	
2	Develop and fit regression models that adjust for correlated data and censored observations	40	x	x	x
3	Interpret regression models and results in public health	20	x	x	x

A1: Attitude

Develop an attitude of discovery/innovation/creativity, as demonstrated by students possessing a strong sense of curiosity, asking questions actively, challenging assumptions or engaging in inquiry together with teachers.

A2: Ability

Develop the ability/skill needed to discover/innovate/create, as demonstrated by students possessing critical thinking skills to assess ideas, acquiring research skills, synthesizing knowledge across disciplines or applying academic knowledge to real-life problems.

A3: Accomplishments

Demonstrate accomplishment of discovery/innovation/creativity through producing /constructing creative works/new artefacts, effective solutions to real-life problems or new processes.

Learning and Teaching Activities (LTAs)

LTAs		Brief Description	CILO No.	Hours/week (if applicable)
1	Teaching	Learning through teaching based on lectures	1, 2, 3	3 hours/ week
2	Assignments	Learning through assignments allows students to perform critical problem analysis and develop hands-on skills involving regression modelling	1, 2, 3	

Assessment Tasks / Activities (ATs)

ATs		CILO No.	Weighting (%)	Remarks ("- " for nil entry)	Allow Use of GenAI?
1	Assignments	1, 2, 3	20	-	No
2	Midterm/quiz	1, 2, 3	40	-	No

Continuous Assessment (%)

60

Examination (%)

40

Examination Duration (Hours)

2

Minimum Continuous Assessment Passing Requirement (%)

40

Minimum Examination Passing Requirement (%)

35

Additional Information for ATs

To pass the course, students are required to obtain a minimum of 40% in continuous assessment and a minimum of 35% in examination.

Assessment Rubrics (AR)

Assessment Task

Assignments (for students admitted before Semester A 2022/23 and in Semester A 2024/25 & thereafter)

Criterion

Problem solving skills

Excellent

(A+, A, A-) Consistently demonstrates a thorough understanding of regression analyses concepts and applies them to complex problems

Good

(B+, B, B-) Adequately demonstrates an understanding of regression analyses concepts and applies them to moderately complex problems

Fair

(C+, C, C-) Demonstrates some understanding of regression analyses concepts and applies them to simple problems

Marginal

(D) Demonstrates some understanding of regression analyses concepts but is unable to apply them to problems

Failure

(F) Demonstrates no understanding of regression analyses concepts and is unable to apply them to problems

Assessment Task

Quizzes/ Midterm Exam (for students admitted before Semester A 2022/23 and in Semester A 2024/25 & thereafter)

Criterion

Problem solving based on comprehensive understanding

Excellent

(A+, A, A-) Demonstrates a comprehensive understanding of regression analyses concepts and applies them to complex problems

Good

(B+, B, B-) Adequately demonstrates an understanding of regression analyses concepts and applies them to moderately complex problems

Fair

(C+, C, C-) Applies regression analyses concepts and methods to solve simple problems

Marginal

(D) Applies regression analyses concepts and methods to solve simple problems but is unable to interpret results

Failure

(F) Inappropriately or unable to apply regression analyses concepts and methods to solve problems

Assessment Task

Examination (for students admitted before Semester A 2022/23 and in Semester A 2024/25 & thereafter)

Criterion

Problem solving based on comprehensive understanding

Excellent

(A+, A, A-) Consistently demonstrates a comprehensive understanding of regression analyses concepts and applies them to complex problems

Good

(B+, B, B-) Adequately demonstrates an understanding of regression analyses concepts and applies them to moderately complex problems

Fair

(C+, C, C-) Demonstrates some understanding of regression analyses concepts and applies them to simple problems

Marginal

(D) Demonstrates some understanding of regression analyses concepts but is unable to apply them to problems

Failure

(F) Demonstrates no understanding of regression analyses concepts and is unable to apply them to problems

Assessment Task

Assignment (for students admitted from Semester A 2022/23 to Summer Term 2024)

Criterion

Problem solving skills

Excellent

(A+, A, A-) Consistently demonstrates a thorough understanding of regression analyses concepts and applies them to complex problems

Good

(B+, B) Adequately demonstrates an understanding of regression analyses concepts and applies them to moderately complex problems

Marginal

(B-, C+, C) Demonstrates some understanding of regression analyses concepts and applies them to simple problems

Failure

(F) Demonstrates no understanding of regression analyses concepts and is unable to apply them to problems

Assessment Task

Midterm/quizzes (for students admitted from Semester A 2022/23 to Summer Term 2024)

Criterion

Problem solving based on comprehensive understanding

Excellent

(A+, A, A-) Demonstrates a comprehensive understanding of regression analyses concepts and applies them to complex problems

Good

(B+, B) Adequately demonstrates an understanding of regression analyses concepts and applies them to moderately complex problems

Marginal

(B-, C+, C) Applies regression analyses concepts and methods to solve simple problems

Failure

(F) Inappropriately or unable to apply regression analyses concepts and methods to solve problems

Assessment Task

Examination (for students admitted from Semester A 2022/23 to Summer Term 2024)

Criterion

Problem solving based on comprehensive understanding

Excellent

(A+, A, A-) Consistently demonstrates a comprehensive understanding of regression analyses concepts and applies them to complex problems

Good

(B+, B) Adequately demonstrates an understanding of regression analyses concepts and applies them to moderately complex problems

Marginal

(B-, C+, C) Demonstrates some understanding of regression analyses concepts and applies them to simple problems

Failure

(F) Demonstrates no understanding of regression analyses concepts and is unable to apply them to problems

Part III Other Information

Keyword Syllabus

Generalized linear models, correlated observations, censored data, longitudinal data.

Reading List

Compulsory Readings

Title	
1	Analysis of Longitudinal Data (Oxford Statistical Science Series) 2nd Editionby Peter Diggle Patrick Heagerty, K.Y. Liang, and Scott Zeger

Additional Readings

Title	
1	Generalized Linear Models with Examples in R (Springer Texts in Statistics) by Dunn and Smyth
2	Applied Survival Analysis Using R by Dirk F. Moore